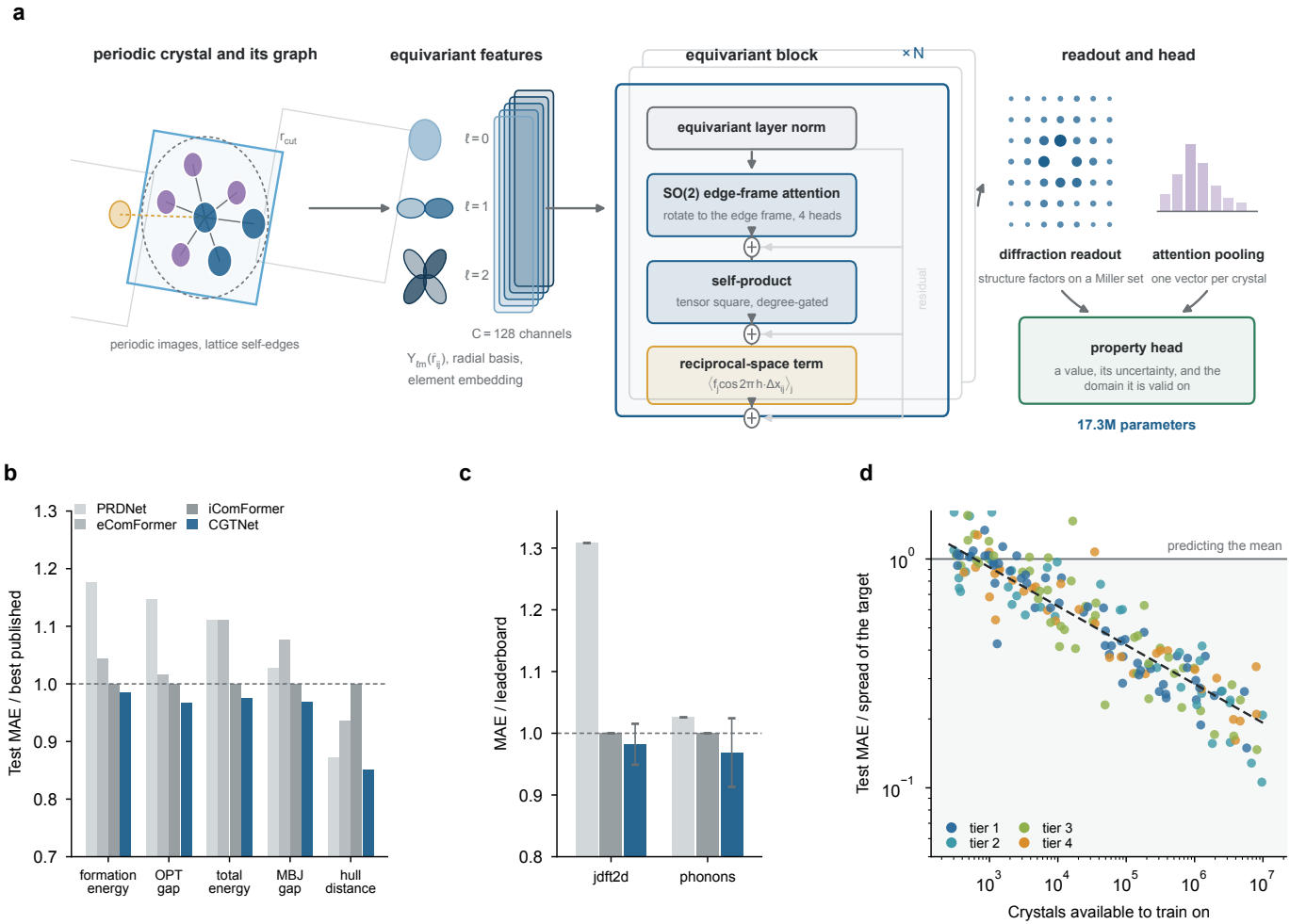

a
periodic crystal and its graph
equivariant features
equivariant block
×N
readout and head
r_{cut}
periodic images, lattice self-edges
ℓ = 0
ℓ = 1
ℓ = 2
C = 128 channels
$Y_{\ell m}(\hat{r}_{ij})$, radial basis,
element embedding
equivariant layer norm
SO(2) edge-frame attention
rotate to the edge frame, 4 heads
self-product
tensor square, degree-gated
reciprocal-space term
$\langle f_i \cos 2\pi\, h \cdot \Delta x_{ij} \rangle_j$
residual
diffraction readout
structure factors on a Miller set
attention pooling
one vector per crystal
property head
a value, its uncertainty, and the
domain it is valid on
17.3M parameters
b
PRDNet
eComFormer
iComFormer
CGTNet
Test MAE / best published
1.3
1.2
1.1
1.0
0.9
0.8
0.7
formation energy
OPT gap
total energy
MBJ gap
hull distance
c
MAE / leaderboard
1.3
1.2
1.1
1.0
0.9
0.8
jdft2d
phonons
d
Test MAE / spread of the target
predicting the mean
10^0
10^{-1}
tier 1
tier 2
tier 3
tier 4
10^3
10^4
10^5
10^6
10^7
Crystals available to train on